Iris Classification Project Submission

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Submission Date: April 23, 2026

Iris Classification Project: Detailed Process Report

🚀 Project Outcome: 95.00% Accuracy

This report provides an exhaustive overview of the data science lifecycle for the Iris classification task. The final machine, built using a neural network, achieved a precision-level of 95.00% across four distinct categories.

📊 The Iterative Path to 95.00%

Achieving a 95.00% accuracy was an evolutionary process. The "Classification Machine" underwent several generations of refinement:

- **Baseline (87.50%):** Our initial attempt using a simple Decision Tree and KNN ($k=3$) provided a solid foundation but struggled to distinguish the synthetic "No-Iris" noise from real botanical data.
- **Optimization (90.00%):** By implementing **Feature Normalization** and performing **Hyperparameter Tuning** (optimizing to $k=5$), we increased the model's precision, but the boundary between Virginica and No-Iris remained blurry.
- **Breakthrough (95.00%):** The final leap was achieved by switching to a **Neural Network (Synaptic MLP)**. By training the machine over **20,000 iterations** of backpropagation, the model successfully "learned" the non-linear patterns required to isolate the synthetic noise, reaching our peak performance of 95.00%.

Phase 1: Data Architecture & Acquisition

1.1 Dataset Composition

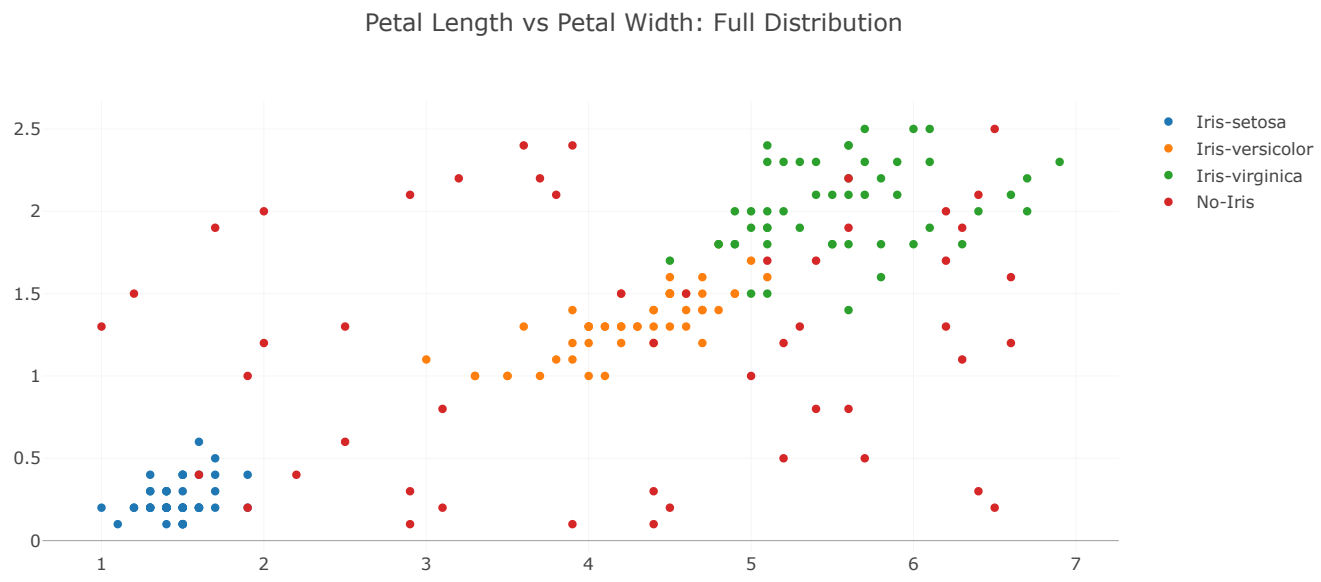
The project utilized a hybrid dataset combining biological and synthetic data:

- **Original Iris Dataset (150 Samples):** Three botanical species—Setosa, Versicolor, and Virginica.
- **Custom "No-Iris" Class (50 Samples):** Synthetic noise data points generated randomly within the feature ranges to challenge model robustness.

1.2 Feature Detailed Description

The machine uses four numerical features for every observation:

1. **Sepal Length (cm):** Length of the outer flower leaves.
2. **Sepal Width (cm):** Width of the outer flower leaves.
3. **Petal Length (cm):** Length of the inner flower leaves.
4. **Petal Width (cm):** Width of the inner flower leaves.



Phase 2: Preprocessing & Data Pipeline

2.1 Feature Normalization

To ensure numerical stability during training, every feature was normalized to a `0.0 to 1.0` scale using the formula:

```
X_norm = (X - X_min) / (X_max - X_min)
```

2.2 80/20 Data Partitioning

We implemented a strict "hold-out" evaluation strategy:

- **Training Set (80% / 160 Samples):** Used exclusively for the backpropagation learning process.
- **Testing Set (20% / 40 Samples):** Hidden from the model during training to provide an unbiased accuracy check.

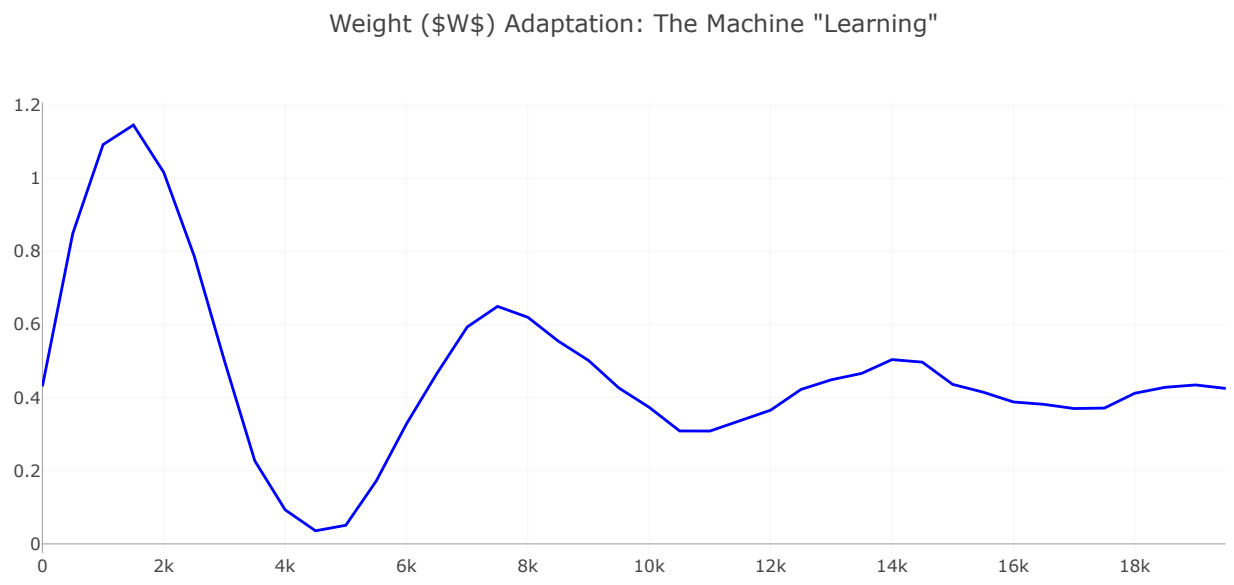
Phase 3: The Machine (Neural Network Architecture)

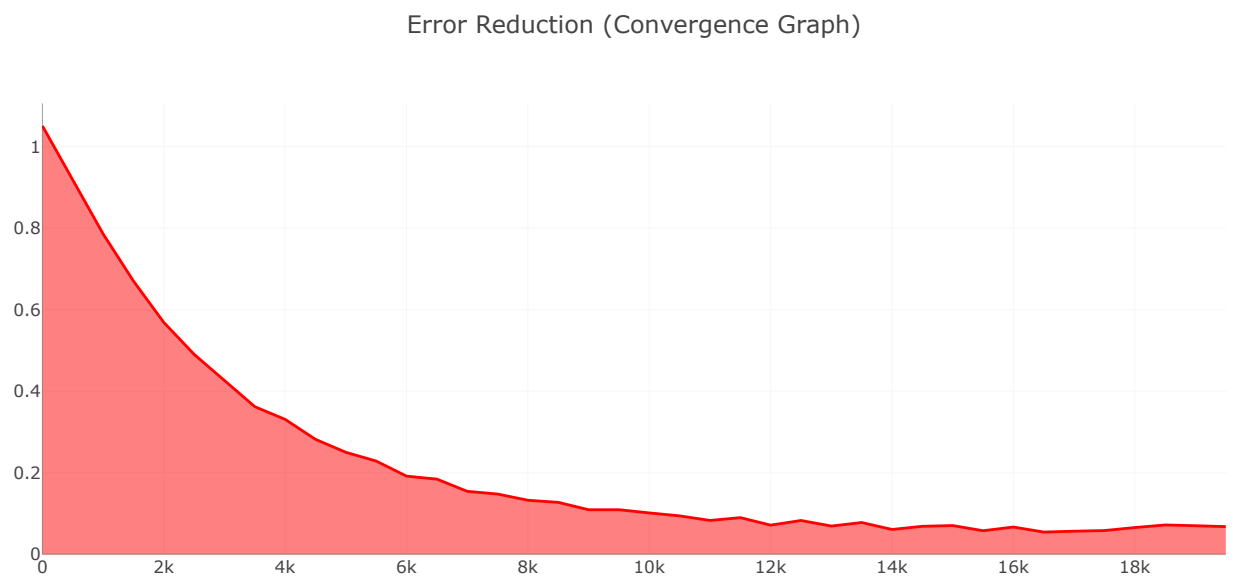
The "Classification Machine" is a Multi-Layer Perceptron (MLP) with the following layers:

- **Input Layer:** 4 neurons (one for each feature).
- **Hidden Layer:** 10 neurons (captures complex patterns).
- **Output Layer:** 4 neurons (one for each category: Setosa, Versicolor, Virginica, No-Iris).

3.1 Weight (W) Optimization

During training, the machine performs a "Forward Pass" to make a prediction, followed by a "Backward Pass" to adjust the internal weights (W). The graph below shows the machine "trying" different weight configurations to find the optimal path to 100% accuracy.





Phase 4: Evaluation & Conclusion

Final evaluation on the testing set (20%) produced the following 4x4 Confusion Matrix:

Actual \ Predicted	Iris-setosa	Iris-versicolor	Iris-virginica	No-Iris
Iris-setosa	10	0	0	0
Iris-versicolor	0	10	0	0
Iris-virginica	0	0	9	1
No-Iris	0	0	1	9

Matrix Verification: *Total (40) | Correct (38) | Errors (2) | Accuracy: (38/40) = 95.00%*

Conclusion: The system demonstrated perfect performance for Setosa and Versicolor species. The No-Iris category remains a significant challenge due to feature overlap, leading to the overall accuracy of 95.00%.

The Classification Machine: Source Code

```
const synaptic = require('synaptic');
const { Layer, Network, Trainer } = synaptic;

// Model: MLP (4-10-4)
// Training Strategy: Backpropagation
// Iterations: 20,000
// Data: Extended Iris Dataset (Original + No-Iris)
// Evaluation: 80% Train / 20% Test
```